Ryan Nelson

ryan@nelsonr.dev
nelsonr.dev
nelson-ryan
ryan-nelson-b3245750

Linguist | ML Engineer

Formal linguist and data engineer
turning toward NLP and machine learning

Education

Master of Science (expected 2026)
University of Arizona
Human Language Technology

Master of Arts
University of Arizona
Linguistics

Bachelor of Arts
University of Arizona
Linguistics and French

Certificate

Natural Language Processing
University of Arizona

Publication

"A principled account of Malagasy deverbal nouns" presented at The 29th West Coast Conference on Formal Linguistics: University of Arizona. April 22, 2011.

Published in Proceedings of the 29th West Coast Conference on Formal Linguistics, ed. Jaehoon Choi et al., 196-204. Somerville, MA: Cascadilla Proceedings Project, 2012.

Work Experience

Topic Modeling Intern
BERTopic · HDBSCAN · UMAP · spaCy · mlflow

2026–current Davis Selected Advisers

DIRTopic
Built a temporal topic model that tracks each topic's prominence against its own baseline using log-odds ratios, visualized as a heat map.

Database Administrator
PowerShell · SQL

2022–2024 Davis Selected Advisers

Analyzed and enhanced existing procedures and scripts for robustness, flexibility, and adaptability.

Established and maintained a centralized PowerShell library of functions and classes for Davis-internal scripts.

Contract Analyst, Various
Python · R · dplyr · JavaScript

2020–2022 Intuit Inc | Celestica Inc | Fiji Water

Deployed automated scripting to streamline reporting and data pipelines across diverse supply chain and operations environments.

Financial Data Analyst
VBA · SQL · R · dplyr

2016–2020 CyraCom International

Owned analysis and reporting of financial and operational business metrics, introducing programmatic procedures to fill reporting gaps and bolster reliability, accuracy, and efficiency.

Other Projects

tagger
Python · Keras · TensorFlow · Weights & Biases

Trains a bigram-convolutional neural network over SpaCy embeddings to predict emotion labels on Reddit comments.

The network design was motivated through evaluation of various models and systematic hyperparameter optimization.

convector
Python · Gensim

Applies distributional semantics and semantic matching toward an automated solver for the New York Times Games Connections puzzle.

Trains CBOW embeddings on Wikipedia and clusters the context vectors via DBSCAN to identify distinct meanings before grouping different words' senses by cosine similarity.

asofterworld
Python · OpenCV · BeautifulSoup

Bridges syntactic theory and NLP to explore whether panel divisions in the webcomic A Softer World separate syntactic constituents.

It scrapes the webcomic's images then detects and partitions panels, assigning OCR-extracted text according to panel boundaries, and passes that text to an NLP pipeline.